INCH-POUND

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SUPERSEDING
MIL-T-25456B (AV)
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COMMERCIAL ITEM DESCRIPTION

TESTER, HOSE ASSEMBLIES, HYDRAULIC-PNEUMATIC, AIRCRAFT AND MISSILE, HYDROCARBON FUEL AND OIL

The General Services Administration has authorized the use of this commercial item description (CID) as a replacement for MIL-T-25456 for all federal agencies.

1. SCOPE

1.1 Scope. This CID describes a test device for conducting proof pressure and leakage tests on fuel and oil hose assemblies for aircraft and missiles.

2. SALIENT CHARACTERISTICS

2.1 Performance. The tester shall be capable of boosting the hydraulic pressure at a minimum ratio of 150 (hydraulic) to 1 (air). It shall have an input air pressure of 100 psi and be capable of producing a maximum hydraulic pressure of 15,000 psi. The tester shall operate satisfactorily with no indication of external leakage when subjected to proof pressure, leakage, and high pressure tests (see 4.3).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to Department of the Army, US Army Aviation and Troop Command, ATTN: AMSAT-R-EDS, 4300 Goodfellow Blvd., St. Louis, MO 63120-1798.

AMSA N/A

FSC
4940

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2.2 Design. All high pressure components, exclusive of gauges, shall be capable of withstanding a pressure of one and a half times the maximum operating pressure of 15,000 psi. The design shall include means of fastening the tester to a work platform.

2.3 Hydraulic components.

2.3.1 Adapters. Suitable adapters shall be furnished with the tester to permit connecting hose assemblies conforming to MS28741-3 to -16 inclusive and MS28762-3 to -16 inclusive, to the hydraulic fluid ports. These adapters shall be stored in racks provided on each side of the unit.

2.3.2 Hydraulic filter. A five micron, or better, hydraulic filter shall be included in the unit between the reservoir and suction side of the pump or within the reservoir. The filter shall be easily removable and replaceable by operator personnel.

2.3.3 Fluid. The unit shall be designed for use with hydraulic fluid conforming to the requirements of MIL-H-5606, MIL -H-82382, or MIL-H-6083.

2.3.4 Gauges. The tester shall be equipped with two hydraulic pressure gauges and one air pressure gauge. The hydraulic pressure gauges shall measure: (1) 0 to 3,000 psig in 25 psi increments and (2) 0 to 15,000 psig in 100 psi increments. They shall have an ASME B40.1 accuracy grade of "A" for both gauges. The air pressure gauge shall measure 0 to 160 psig in 10 psi increments. The ASME B40.1 accuracy grade shall be "D" or better. A gauge saver shall be incorporated before the low pressure gauge to protect it when pressures exceed the maximum range of the gauge.

2.3.5 Sight tube. Means shall be provided to fill the test hose under low pressure. This shall include provisions for bleeding all air out of the hydraulic test circuit and a sight tube to permit visual inspection of the bleeding operation. The sight tube shall be positioned so that it does not siphon air bubbles from the discharge side. The sight tube shall not be subjected to operating pressures.

2.3.6 Reservoir. The reservoir shall have a capacity of five gallons, plus an expansion volume of at least .5 gallons above the full level. If the top of the reservoir is not removable, a clean-out hole at least five inches in diameter and a drain valve shall be furnished. A means of filtering replacement fluid with at least a 30 mesh strainer shall be provided. Also, the reservoir shall be provided with a means to indicate a full level.

2.3.7 Valves. A bleed valve shall be provided to permit draining of the oil out of the test hose back into the reservoir after completion of the test.

2.4 Air components.

2.4.1 Pump. The air operated pump shall have an oil cycle capacity of 0.30 gpm when operating with 100 psi of supply air at 8 cfm.

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2.4.2 Air filter. A 10 micron air filter shall be provided at the air inlet port. The air filter shall be equipped with a moisture trap, glass or plastic sight bowl, and drain valve.

2.4.3 Pressure regulator. An adjustable air regulator valve shall be provided to limit the hydraulic outlet pressure between 300 and 15,000 psi.

2.4.4 Lubrication. Means shall be provided to automatically lubricate the pump.

2.4.5 Muffler. A suitable muffler shall be provided which will reduce the noise of the pump to a level not to exceed 80 decibels at a distance of five feet in any direction from the tester.

2.4.6 Air shut-off valve. A shut-off valve shall be provided in the air line to the pump to shut off air flow to the pump.

2.5 Interchangeability. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable.

2.6 Finish. All parts except glass shall be painted, anodized, or finished in accordance with best commercial practices, as appropriate for the type of material.

3. QUALITY ASSURANCE PROVISIONS

3.1 Contractor certification. The contractor shall certify and maintain substantiating evidence that the product offered meets the salient characteristics of this commercial item description, and that the product conforms to the producer’s own drawings, specifications, standards, and quality assurance practices, and is the same product offered for sale in the commercial marketplace. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

3.2 Test conditions. Unless otherwise noted, ambient test conditions shall be as specified herein:

Temperature	-5°F and 125°F
Pressure	site pressure
relative humidity	uncontrolled room ambient

3.3 Individual tests. Two testers will be each temperature conditioned as above and then subjected to the following tests:

a. Functional test
b. High pressure test
c. Gauge test

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Failure of any tests shall be cause for rejection of that particular unit.

3.3.1 Functional test. Using a tester as the pressure source, a proof pressure test and a leakage test shall be conducted as outlined in table I. Tests shall be conducted in accordance with FED-STD-601, method 10211. Leakage at any point, failure of any component, or a reduction of more than one percent of specified test pressure shall not be acceptable.

TABLE I
PROOF PRESSURE AND LEAKAGE TESTS

Hose Assembly	Proof Pressure (psi)	Leakage (psi)	Time Application of Pressure (min)
MS28741-4-0120	6000	8400	5
MS28741-12-0120	3000	4200	5
MS28759-4-0120	8000	1120	3 to 5
MS28759-12-0120	6000	8400	3 to 5

3.3.2 High pressure test. The hydraulic system, with the exception of the gauges, shall be subjected to a pressure of 22,500 psi for a period of five minutes. There shall be no leakage or failure of any components. Tests shall be conducted in accordance with FED STD 601, method 10211.

3.3.3 Gauge test. Each gauge shall be calibrated at full scale against a master gauge or tester with an inherent accuracy of .25 of one percent. Each gauge shall then be cycled from 0 to full scale and back to 0, 10 times. Each gauge shall then be rechecked against the master gauge at full scale. The initial calibration shall not vary by more than one percent.

4. PACKAGING.

4.1 Preservation-packaging. Preservation, packing, and marking shall be as specified in the contract or order.

5. NOTES

5.1 Applicable documents. Copies of specifications, MS sheets, and standards referenced in this CID are available from the Standardization Document Order Desk, 700 Robbins Avenue, Bldg. 4D, Philadelphia, PA 19111-5094.

Copies of ASME publications may be obtained from American Society of Mechanical Engineers, 345 E. 47th St., New York, NY 10017.

5.2. Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this CID

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b. Level of preservation and packaging and packing required.
c. Whether special markings for shipment are required.

5.3 National stock number. NSN: 4940-00-855-9035

5.4 Commercial products. A tester meeting the requirements of this CID is manufactured by Teledyne Fluid Systems, Sprague Products Division, (216-267-2700), item No. S439-JR-150, (for information purposes only).

CONCLUDING MATERIAL

CUSTODIAN:
Army - AV
Navy - AS
Air Force - 99

PREPARING ACTIVITY:
Army - AV

Project No. 4940-0677

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